

EEE4114F: Problem Set 6 (LTIF)

1. When the input to a linear time-invariant system is

$$x[n] = \left(\frac{1}{2}\right)^n u[n] + (2)^n u[-n-1]$$

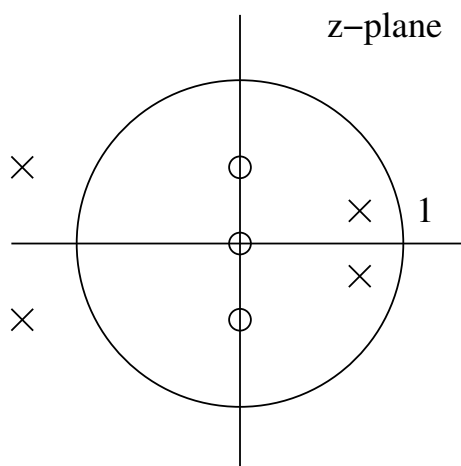
the output is:

$$y[n] = 6 \left(\frac{1}{2}\right)^n u[n] - 6 \left(\frac{3}{4}\right)^n u[n].$$

- (a) Find the system function $H(z)$ of the system. Plot the poles and zeros of $H(z)$, and indicate the region of convergence.
 - (b) Find the impulse response $h[n]$ of the system for all values of n .
 - (c) Write the difference equation that characterises the system.
 - (d) Is the system stable? Is it causal?
2. Consider a system described by a linear constant-coefficient difference equation with initial-rest conditions. The step response of the system is given by

$$y[n] = \left(\frac{1}{3}\right)^n u[n] + \left(\frac{1}{4}\right)^n u[n] + u[n].$$

- (a) Determine the difference equation.
 - (b) Determine the impulse response of the system.
 - (c) Determine whether or not the system is stable.
3. The system function of a linear time-invariant system has the pole-zero plot shown below:



Specify whether each of the following statements is true, is false, or cannot be determined from the information given.

- (a) The system is stable.
- (b) The system is causal.
- (c) If the system is causal, then it must be stable.
- (d) If the system is stable, then it must have a two-sided impulse response.

4. For each of the following system functions, state whether or not it is a minimum-phase system. Justify your answers:

$$H_1(z) = \frac{(1 - 2z^{-1})(1 + \frac{1}{2}z^{-1})}{(1 - \frac{1}{3}z^{-1})(1 + \frac{1}{3}z^{-1})}$$

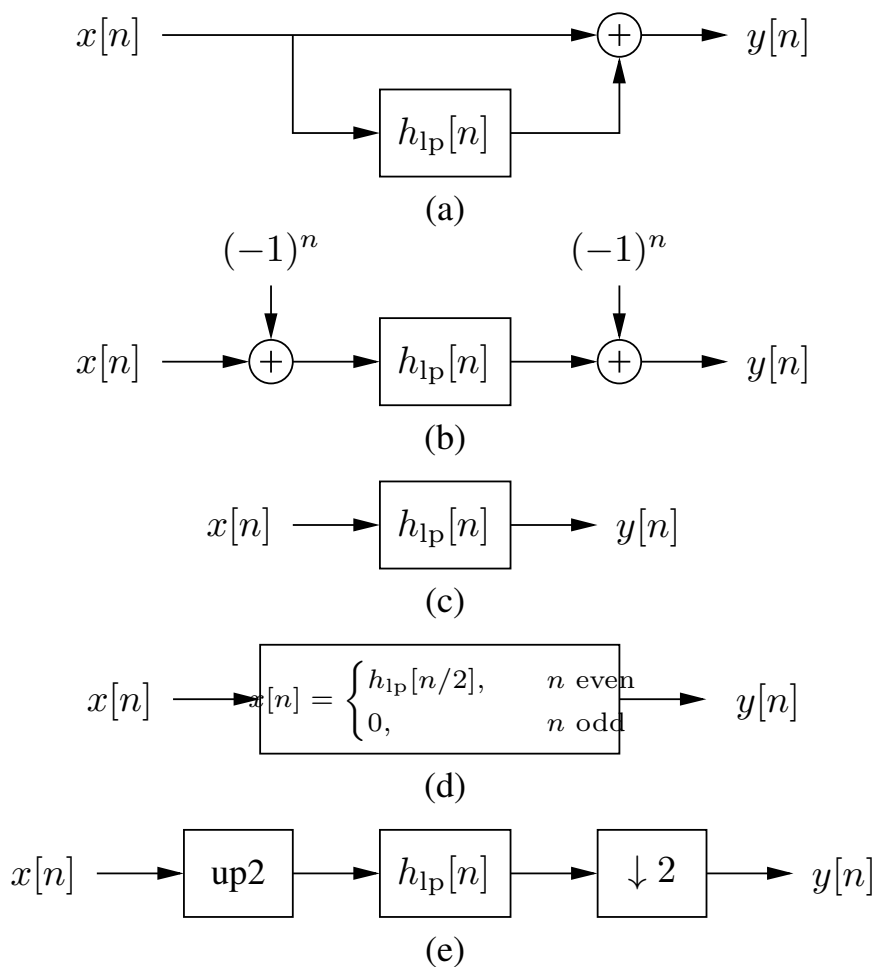
$$H_2(z) = \frac{(1 + \frac{1}{4}z^{-1})(1 - \frac{1}{4}z^{-1})}{(1 - \frac{2}{3}z^{-1})(1 + \frac{2}{3}z^{-1})}$$

$$H_3(z) = \frac{1 - \frac{1}{3}z^{-1}}{(1 - \frac{j}{2}z^{-1})(1 + \frac{j}{2}z^{-1})}$$

$$H_4(z) = \frac{z^{-1}(1 - \frac{1}{3}z^{-1})}{(1 - \frac{j}{2}z^{-1})(1 + \frac{j}{2}z^{-1})}$$

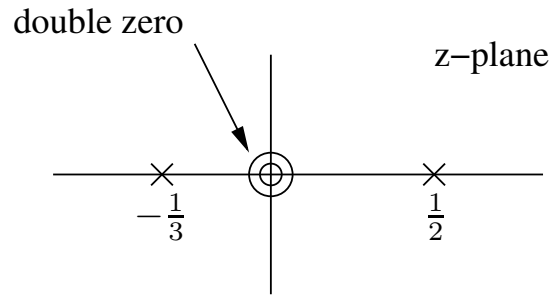
5. Let $h_{lp}[n]$ denote the impulse response of an ideal lowpass filter with unity passband gain and cutoff frequency $\omega_c = \pi/4$. The figure

below shows five systems, each of which is equivalent to an ideal LTI frequency-selective filter:



For each system shown, sketch the equivalent frequency response, indicating explicitly the band-edge frequencies in terms of ω_c . In each case, specify whether the system is a lowpass, bandpass, highpass, bandstop or multiband filter.

6. The system function $H(z)$ of a causal linear time-invariant system has the pole-zero configuration shown below:



It is also known that $H(z) = 6$ when $z = 1$.

- (a) Determine $H(z)$.
- (b) Determine the impulse response $h[n]$ of the system.
- (c) Determine the response of the system to the following input signals:
 - i. $x[n] = u[n] - \frac{1}{2}u[n-1]$
 - ii. The sequence $x[n]$ obtained from sampling the continuous-time signal

$$x(t) = 50 + 10 \cos(20\pi t) + 30 \cos(40\pi t)$$

at a sampling frequency $\omega_s = 2\pi(40)$ rad/s.

7. The system function H_{II} represents a type of II FIR generalised linear-phase system with impulse response $h_{II}[n]$. This system is cascaded with an LTI system whose system function is $(1 - z^{-1})$ to produce a third system with system function $H(z)$ and impulse response $h[n]$. Prove that the overall system is a generalised linear-phase system, and determine what type of linear-phase system it is.
8. A causal linear shift-invariant system has a system function

$$H(z) = \frac{1 + z^{-1}}{1 - \frac{1}{2}z^{-1}}.$$

Find the z-transform of the input, $x[n]$, that will produce the output

$$y[n] = -\frac{1}{3} \left(\frac{1}{4}\right)^n u[n] - \frac{4}{3}(2)^n u[-n-1].$$

9. A signal $y[n]$ contains a primary signal, $x[n]$, plus two echos:

$$y[n] = x[n] + \frac{1}{2}x[n - n_d] + \frac{1}{4}x[n - 2n_d].$$

Find a realizable filter that will recover $x[n]$ from $y[n]$.

10. A nonminimum phase causal sequence $x[n]$ has a z-transform

$$X(z) = \frac{(1 - \frac{3}{2}z^{-1})(1 + \frac{1}{3}z^{-1})(1 + \frac{5}{3}z^{-1})}{(1 - z^{-1})^2(1 - \frac{1}{4}z^{-1})}.$$

For what values of a constant α will the sequence $y[n] = \alpha^n x[n]$ be minimum phase?

11. We wish to design an FIR lowpass filter satisfying the specifications

$$0.95 < H(e^{j\omega}) < 1.05, 0 \leq |\omega| \leq 0.25\pi,$$

$$-0.1 < H(e^{j\omega}) < 0.1, 0.35\pi \leq |\omega| \leq \pi,$$

by applying a window $w[n]$ to the impulse response $h_d[n]$ for the ideal discrete-time lowpass filter with cutoff $\omega_c = 0.3\pi$. Which of the filters listed below can be used to meet this specification? For each window that you claim will satisfy this specification, give the minimum length $M + 1$ required for the filter.

Rectangular

$$w[n] = \begin{cases} 1 & 0 \leq n \leq M \\ 0 & \text{otherwise} \end{cases}$$

Bartlett (triangular)

$$w[n] = \begin{cases} 2n/M & 0 \leq n \leq M/2 \\ 2 - 2n/M & M/2 < n \leq M \\ 0 & \text{otherwise} \end{cases}$$

Hanning

$$w[n] = \begin{cases} 0.5 - 0.5\cos(2\pi n/M) & 0 \leq n \leq M \\ 0 & \text{otherwise} \end{cases}$$

Hamming

$$w[n] = \begin{cases} 0.54 - 0.46\cos(2\pi n/M) & 0 \leq n \leq M \\ 0 & \text{otherwise} \end{cases}$$

Blackman

$$w[n] = \begin{cases} 0.42 - 0.5 \cos(2\pi n/M) + 0.08 \cos(4\pi n/M) & 0 \leq n \leq M \\ 0 & \text{otherwise} \end{cases}$$

12. The commonly used windows presented in the previous question can all be expressed in terms of rectangular windows. This fact can be used to obtain expressions for the Fourier transforms of the Bartlett window and the raised-cosine family of windows, which includes the Hanning, Hamming and Blackman windows.

- (a) Show that the $(M + 1)$ -point Bartlett window, defined previously, can be expressed as the convolution of two smaller rectangular windows. Use this fact to show that the Fourier transform of $(M + 1)$ -point Bartlett window is

$$W_B(e^{j\omega}) = e^{-j\omega M/2} (2/M) \left(\frac{\sin(\omega M/4)}{\sin(\omega/2)} \right)^2,$$

for M even or

$$W_B(e^{j\omega}) = e^{-j\omega M/2} (2/M) \left(\frac{\sin[\omega(M+1)/4]}{\sin(\omega/2)} \right) \left(\frac{\sin[\omega(M-1)/4]}{\sin(\omega/2)} \right),$$

for M odd.

- (b) It can easily be seen that the $(M+1)$ -point raised-cosine windows (Hanning, Hamming and Blackman) can all be expressed in the form

$$w[n] = [A + B \cos(2\pi n/M) + C \cos(4\pi n/M)] w_R[n]$$

where $w_R[n]$ is an $(M+1)$ -point rectangular window. Use this relation to find the Fourier transform of the general raised-cosine window.

- (c) Using appropriate choices for A, B and C and the result determined in Part (b), sketch the magnitude of the Fourier transform of the Hanning window.

13. Derive

$$H(e^{j\omega}) = e^{-jN\omega/2} \sum_{k=1}^{(N+1)/2} b(k) \cos \left[\left(k - \frac{1}{2} \right) \omega \right]$$

where

$$b(k) = 2h \left(\frac{N+1}{2} - k \right)$$

$$k = 1, 2, \dots, \frac{N+1}{2}$$

for the frequency response of a type II linear phase filter.